Cristian Bautista

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SUMMARY

Ph.D. student in Electrical and Computer Engineering at The Ohio State University with experience in label-free and collaborative 3D perception and full-stack autonomous vehicle integration. Research interests include learning-based decision-making and Vision-Language-Action models for autonomous driving.

EDUCATION

The Ohio State University, May 2026 – Expected May 2030
Ph.D. in Electrical and Computer Engineering GPA:
Advisors: Qadeer Ahmed, Ekim Yurtsever

The Ohio State University, Aug 2024 – May 2026
M.S. in Electrical and Computer Engineering GPA: 3.56 / 4.0
Selected Courses: Reinforcement Learning (A), Autonomy in Vehicles (A), Robotics (A), Online Learning & Optimization (A-)
Advisors: Wei-Lun Chao, Qadeer Ahmed

Universidad Santo Tomás, Bogotá, Colombia Feb 2016 – Jun 2021
B.S. in Electronic Engineering GPA: 4.4 / 5.0
Minor in Computer Vision and Machine Learning
Graduated with Highest Honors

SELECTED PUBLICATIONS

Zhen Xu*, Jinsu Yoo*, **Cristian Bautista***, Zanming Huang, Tai-Yu Pan, Zhenzhen Liu, Katie Z. Luo, Mark Campbell, Bharath Hariharan, and Wei-Lun Chao. *"When the City Teaches the Car: Label-Free 3D Perception from Infrastructure."* European Conference on Computer Vision (**ECCV**), 2026; CVPR DriveX Workshop, **Best Poster Award**.

Cristian Bautista, Andrei Kopanev, Qizhe Xu, Xinyu Ma, Jiyao Yuan, Sooyoung Jeon, Derin Durak, Tian-Le Dai, Yuxin Xiao, Rocky Shao, Shengzhe Tan, Gowrav Mannem, Rayaan Qureshi, Jinsu Yoo, Lisa Fiorentini, Wei-Lun Chao, and Qadeer Ahmed. *"BRUTUS: A Full-Stack Autonomous Driving Platform in the Robotaxi Era."* In Submission for the *Journal of Field Robotics*, 2026.

Zhen Xu*, **Cristian Bautista***, Jinsu Yoo*, Zanming Huang, and Wei-Lun Chao. *"SmartCityZero: Label-Free 3D Perception through RSU-to-Vehicle Collaboration."* **ICCV** X-Sense Workshop, 2025.

Cristian Bautista*, Yesid Fonseca*, Camilo Pardo, and Carlos Parra. *"A Plum Selection System that Uses a Multiclass Convolutional Neural Network." Journal of Agriculture and Food Research*, 2023.

*Equal contribution.

HONORS & AWARDS

2026: **Best Poster Award** — *CVPR DriveX Workshop*

2025–2026: **SAE AutoDrive Challenge II** — *Buckeye AutoDrive*

- **Two-time 3rd Place**, Mobility Innovation (2025, 2026) (Leader & Presenter)
- **2nd Place**, MathWorks Simulation Challenge (2026) (Team Captain)
- **3rd Place**, Overall Static Events (2026) (Team Captain)
- **3rd Place**, System Safety (2026) (Team Captain)

2023: Recipient of the **COLFUTURO Scholarship** — *Colombian Ministry of Science, Technology and Innovation*

2022: Recipient of the **Opportunity Funds Program** — *U.S. Department of State*

2021: Highest Honors Graduation **Cum Laude and Laureate Thesis Award** — *Universidad Santo Tomás*

2018–2019: Recipient of a **Merit-Based Scholarship** — *Universidad Santo Tomás*

2018: Recipient of an **International Exchange Scholarship** — *Universidad Santo Tomás*

RESEARCH EXPERIENCE

The Ohio State University **Columbus, OH**
Graduate Research Associate *Summer 2025 – Present*

- **Label-Free and Collaborative 3D Perception:** Developed infrastructure-to-vehicle collaboration methods for label-free 3D object detection, leveraging roadside sensors to improve ego-vehicle perception without manual annotations.
- **Learning-Based Decision-Making:** Designed hierarchical deep reinforcement learning agents for decision-making and path planning in simulated urban robotaxi missions.

Buckeye AutoDrive, The Ohio State University **Columbus, OH**
Team Captain (Oct. 2025–Jun. 2026) & Mobility Innovation Lead *Aug. 2024 – Jun. 2026*

- Led a multidisciplinary team of 40+ students in the development, integration, and real-world deployment of BRUTUS, a full-stack autonomous research vehicle.
- Led end-to-end integration across multimodal perception, localization, path planning, control, distributed computing and CAN-based vehicle interfaces.
- Developed MILP-based sensor-suite optimization and collaborative-driving concepts to mitigate misdirection from traffic agents, earning 3rd Place in Mobility Innovation in 2025; Oral presentation at SAE WCX 2025.
- Led the development of a Vision-Language-Action concept to improve passenger experience and AV transparency, earning 3rd Place in Mobility Innovation in 2026.

Universidad Santo Tomás **Tunja, Colombia**
Plum Selection System — Undergraduate Thesis *2019–2021*

- Developed a multiclass CNN to grade plums using morphological and visual features, incorporating data augmentation and Grad-CAM for model analysis and interpretability.
- Integrated the trained model into a real-world robotic sorting system for automated fruit classification and handling.

INDUSTRY EXPERIENCE

Cummins Inc. & Center for Automotive Research **Columbus, OH**
Automotive Embedded Systems Research Project *Summer 2025*

- Analyzed AUTOSAR-based embedded software architectures to support alignment with automotive functional-safety requirements.
- Mapped ISO 26262 safety requirements to verification artifacts and supported end-to-end requirements traceability through application lifecycle management (ALM) workflows.

Technip Energies **Bogotá, Colombia**
Instrumentation & Automation Specialist I *2022–2024*

- Led the specification, procurement, and integration of pressure and temperature instrumentation for a European biofuels plant.
- Coordinated with global suppliers to evaluate technical proposals and validate industrial instrumentation against process and project requirements.
- Produced process and pneumatic hook-up diagrams for field instrumentation and instrument-air distribution systems.

TECHNICAL SKILLS

Programming & Systems: Python, C++, MATLAB, Bash, Linux, Git, Slurm, GPU/CPU computing

Software & Libraries: PyTorch, TensorFlow, ROS 2, OpenCV, Open3D, CUDA, CARLA

Autonomous Systems: 2D/3D perception, Multi-agent perception, End-to-end integration

Topics: Reinforcement learning, Optimization, Machine Learning, Unsupervised 3D Detection, Vision-language-action models, Online Learning (Multi-arm bandits, UCB, EXP3)

Languages: English (full professional proficiency), Spanish (Native)

TEACHING EXPERIENCE

Universidad Santo Tomás **Tunja, Colombia**
Teaching Assistant, Fourier Analysis *Fall 2019 – Spring 2020*

- Delivered lectures, held office hours, and supported laboratory sessions for 50+ undergraduate students, covering Fourier series, Fourier transforms, and signal-processing applications.

PROFESSIONAL SERVICE

2026: Reviewer, *DriveX Workshop at ECCV 2026*

2026: Reviewer, *23rd IFAC World Congress*